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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING		
ACADEMIC YEAR: 2026-27	**BCS586 MINI PROJECT**	**SEM V**

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DOMAIN	**Networking**	
TITLE (proposed)	**NetShield: Machine Learning-Based Intrusion Detection System**	
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Title: NetShield: Machine Learning-Based Intrusion Detection System

Problem Statement

The problem objective is to design and develop an intelligent and automated Network Intrusion Detection System using machine learning techniques to identify abnormal and potentially malicious network traffic, in order to overcome the limitations of traditional signature-based detection systems, improve detection accuracy, reduce false alarms, and support efficient real-time network security monitoring.

Abstract

Traditional intrusion detection systems mainly depend on predefined attack signatures, making it difficult to detect new and evolving threats. This project proposes an AI-powered Network Intrusion Detection System that uses machine learning to classify network traffic as normal or anomalous. The system performs data preprocessing, feature selection, and feature transformation before applying classification algorithms such as Logistic Regression, AdaBoost, Random Forest, XGBoost, and SVM. The models are evaluated using accuracy, precision, recall, F1-score, and confusion matrix. The proposed system provides an automated approach for detecting suspicious network activity and can be further extended for live packet capture and real-time network monitoring.

Functional Requirements

- Collect and process network traffic datasets.
- Preprocess and clean network traffic data.
- Select relevant features for intrusion detection.
- Train multiple machine learning classification models.
- Classify network traffic as normal or anomalous.
- Compare model performance using standard evaluation metrics.
- Display intrusion detection results and network activity.
- Generate performance reports and detection statistics.

Non-Functional Requirements

- **Accuracy** in detecting abnormal network traffic.
- **Scalability** for handling large network traffic datasets.
- **Reliability** in producing consistent detection results.
- **Efficiency** with low model inference and detection time.
- **Maintainability** through a modular system architecture.
- **Usability** through a simple monitoring and visualization interface.
- **Extensibility** for integrating additional models and future real-time monitoring.

Technical Requirements

Software

- Python programming language
- Scikit-learn machine learning library
- Pandas and NumPy for data processing
- Matplotlib / Seaborn for data visualization
- Flask / FastAPI for backend development
- React.js / HTML, CSS, JavaScript for dashboard
- SQLite / PostgreSQL database
- Jupyter Notebook / VS Code
- Git and GitHub

Hardware

- Computer or laptop with minimum 8 GB RAM
- Multi-core processor
- Minimum 10 GB available storage
- Stable internet connection
- Network interface for network traffic monitoring

Other

- Network traffic dataset
- Preprocessed and labelled training data
- Machine-learning model files
- Network monitoring/packet capture tools
- Web browser for dashboard access
- Dataset and model evaluation tools

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Feasibility Analysis

Technical Feasibility

The system uses machine-learning algorithms and open-source tools for network traffic analysis and intrusion detection. The required technologies can be integrated using standard development frameworks.

Operational Feasibility

The system provides an automated approach to identify suspicious network traffic and display detection results. It can help users monitor network activity and understand potential intrusions.

Economic Feasibility

Since the project can be developed using open-source software, libraries, and publicly available datasets, the implementation cost is minimal. It can be developed using existing computer hardware and basic infrastructure.

Limitations

- Performance depends on the quality and diversity of the dataset.
- Offline testing may not fully represent real-world network conditions.
- New or unseen attacks may not be detected accurately.
- False positives and false negatives may occur.
- Live packet capture and real-time deployment are not fully implemented.
- Model performance may vary across different network environments. IEEE GG

Conclusion

The **AI-Powered Network Intrusion Detection System using Machine Learning** provides a machine-learning-based approach for identifying abnormal network traffic. The proposed system performs data preprocessing, feature selection, model training, traffic classification, and performance evaluation.

Multiple machine-learning algorithms can be evaluated to determine their effectiveness for intrusion detection. The system can also provide visualization and monitoring capabilities to make detection results easier to understand.

The approach is technically and economically feasible for a college minor project because it can be developed using commonly available hardware, open-source software, and machine-learning libraries. However, an important limitation is that an offline dataset cannot completely represent the complexity of live network environments.

In future development, the system can be extended with **live packet capture, real-time traffic monitoring, improved feature engineering, additional attack types, and deployment in real network environments**, which are also identified as future directions in the reference paper.